

LARRY OTIENO

Nairobi, Kenya | otienolarry7@gmail.com | 0790 247 511 | github.com/larry-otieno

PROFESSIONAL SUMMARY

Fourth-year Computer Security and Forensics student at Kabarak University, finishing my undergraduate degree with hands-on full-stack development experience and a primary focus on frontend architecture (Next.js, React, TypeScript). Specializes in building polished, responsive UIs with security engineered in from the ground up — authentication, encryption, and audit logging — shaped by formal training in cybersecurity and digital forensics. Comfortable owning UI/UX design and REST API integration independently, with the security instincts to recognize and close the vulnerabilities that matter most.

TECHNICAL SKILLS

Frontend & UI/UX (primary strength): HTML5, CSS3, JavaScript (ES6+), Next.js (App Router), React, TypeScript, Tailwind CSS, Radix UI/shadcn, TanStack Query, Zustand, React Hook Form + Zod, Recharts, responsive/mobile-first design, REST API integration (Axios).

Backend & APIs: PHP 8 (OOP, PDO), RESTful API design, Python (FastAPI), Node.js (Next.js SSR/API routes), MySQL, PostgreSQL — schema design, migrations, query optimization.

Security & Forensics: Authentication & MFA (email-OTP, TOTP, JWT/RS256, Argon2id), encryption (AES-256-GCM envelope encryption), tamper-evident audit logging (SHA-256/HMAC hash chains), threat modeling, incident response, OWASP ZAP, Burp Suite, forensic tools (Autopsy, FTK).

Systems & Tooling: Git, GitHub Actions CI/CD, Docker, PHPStan, ESLint, cPanel, Windows/Linux administration.

KEY PROJECTS

Oasis of Peace — School & Children's Home Management Platform

Full-Stack Developer (2-person team) | January 2026 – June 2026

Full-stack platform for St. Mary Magdalene Oasis of Peace serving three audiences from one codebase — a public marketing site, a parent portal, and a role-based staff dashboard. Owned the frontend end-to-end and roughly half of the backend.

- Built the entire Next.js 16 / React 19 / TypeScript frontend: 17 feature modules, 52 routes, 26 dashboard pages, and ~200 components, including user-scoped dark mode and cookie-consent-gated GA4 analytics.
- Co-designed a six-role RBAC + ABAC policy engine (31 guarded resources) and implemented the frontend/auth side of email-OTP two-factor authentication, refresh-token rotation, and a CSP with per-request nonces.
- Integrated Paystack donations with server-side verification and idempotent webhooks; built fee/finance dashboards with live parent balance calculations, and set up the GitHub Actions CI/CD pipeline (ESLint, tsc --noEmit, PHPStan, PHP tests) as a merge gate.

Stack: Next.js 16, React 19, TypeScript, Tailwind CSS v4, Radix UI/shadcn, TanStack Query, Zustand, React Hook Form + Zod, Recharts · PHP 8.1 REST API, PDO, MySQL 8

Note: Source code is hosted in a private repository; core architectural schemas can be demonstrated via an obscured reference repository upon request.

Secure Examination Management System (SEMS) — Undergraduate Capstone

Lead Developer & Architect | May 2026 – July 2026

End-to-end document-leakage prevention platform combining a Next.js frontend with a Python security backend, built to stop exam paper leaks via forensic watermarking and strict access control.

- Built the Next.js 15 / React 19 frontend, integrated with a FastAPI watermarking microservice and PostgreSQL/Drizzle backend; implemented client-side auth flows for Argon2id hashing, RS256 JWT (15-min access / 7-day refresh), and mandatory TOTP MFA with secure HttpOnly/Secure/SameSite cookies.
- Enforced a strict secure-download UX chain (time-lock → policy → decrypt → watermark → stream) with forced Content-Disposition and a PDF/DOCX MIME allowlist.

Stack: Next.js 15, React 19, TypeScript, Python 3.11 (FastAPI), PostgreSQL 15, Drizzle ORM, MinIO, Docker

Outcome: 469/469 tests passing, 0 typecheck errors at stable baseline (v0.6.2)

Note: Source code is hosted in a private repository; architecture and packaging pipeline can be demonstrated upon request.

PhishGuard-AI — Phishing URL Detection System

Backend Developer | AI Mini-Project 2026 | Team of 5 | github.com/AI-Mini-Project-for-2026/phishguard-ai | Demo: phishguard-ai-2026.streamlit.app

- Built the backend prediction pipeline (URL normalization, 11 heuristic features, model inference) wired into a Streamlit frontend, with joblib model/scaler persistence and a graceful demo-mode fallback when trained model files are unavailable.

Stack: Python, pandas, scikit-learn, XGBoost, Streamlit, Flask

WORK EXPERIENCE

ICT Intern (Industrial Attachment) — National Gender and Equality Commission (NGEC), ICT Division

October 2025 – December 2025 | Nairobi, Kenya

- Maintained and troubleshooted desktop, network, and peripheral hardware, installing updates and performing routine checks to prevent downtime; provided helpdesk support resolving software, email, and connectivity issues for staff.
- Implemented and tested data backup/recovery procedures for critical records against data protection standards; managed IT asset inventory and device configuration, and worked with SQL Server Management Studio.

EDUCATION

Kabarak University — B.Sc. Computer Security and Forensics, Fourth Year | Expected 2026

Relevant coursework: Cybersecurity Fundamentals, Digital Forensics, Network Security, Database Systems.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Diploma in Cyber Security (NIST CSF 2.0) & Cloud Security (CSA v5) ·
- IBM SkillsBuild Cybersecurity Certificate (In Progress)